

RETAIL TRANSACTIONS DATA ANALYSIS & DASHBOARD DEVELOPMENT

Comprehensive Project Report • Data Analytics Workflow

ABSTRACT

This project focuses on analyzing a massive retail transactions dataset consisting of 1 million records collected between the years 2020 and 2024. The primary objective is to systematically clean the dataset, execute thorough Exploratory Data Analysis (EDA), identify overarching business trends, and construct an interactive Power BI dashboard. The resulting comprehensive analysis yields deep business insights into customer purchasing behaviors, payment method distributions, seasonal trends, store-type performance dynamics, and promotional strategy impacts.

PROJECT OBJECTIVES

To successfully deliver actionable business intelligence, the project was executed against the following structured technical and functional milestones:

- **Data Preprocessing:** Clean and format the complex retail transaction dataset.
- **Data Integrity:** Effectively handle missing structural values and eliminate duplicate records.
- **Exploratory Data Analysis:** Perform multi-dimensional statistical analysis (Univariate, Bivariate, and Multivariate).
- **Behavioral Insight:** Uncover systemic customer purchasing patterns and segment profiles.
- **Operational Strategy:** Study operational variables such as payment types, transaction channels, and promotions.
- **Business Intelligence:** Architect and deploy a production-ready interactive Power BI dashboard.
- **Executive Recommendations:** Transform raw data metrics into strategic insights for operational decision-making.

DATASET DESCRIPTION

The underlying analytics dataset encompasses a granular cross-section of retail transaction telemetry containing 1,000,000 distinct operational logs spanning a four-year period.

Attribute	Functional Description
Transaction_ID	Unique system-generated alphanumeric transaction identifier.
Date	Temporal timestamp noting when the transaction occurred.
Customer_Name	Full name profile of the transacting customer.
Product	Categorized product description or item SKU purchased.
Total_Items	Discrete volumetric count of items bundled within the single transaction.
Total_Cost	Net monetary bill amount denominated in USD, reflecting all items purchased.
Payment_Method	Primary mode of transaction settlement (e.g., Credit Card, Cash, Digital Wallet).
City	Geographic location of the retail branch where the purchase took place.
Store_Type	Categorization of the store footprint (e.g., Supermarket, Convenience Store, Boutique).
Discount_Applied	Boolean flag specifying whether a pricing markdown or coupon was applied.
Customer_Category	Demographic or behavioral loyalty tier categorization of the customer.
Season	Temporal season tag mapped to the transaction date (Winter, Spring, Summer, Autumn).
Promotion	Indicates the specific marketing or promotional campaign active during checkout.

METHODOLOGY

The project adhered strictly to a standard data science lifecycle framework, taking the raw unstructured transactional data through a repeatable pipeline to final visualization.

Step 1: Scalable Data Collection & Ingestion

The primary ingestion layer processed a large-scale data schema delivered in a normalized flat file structure.

1 Million Records

CSV Format

Temporal Horizon: 2020 - 2024

Step 2: Rigorous Data Cleaning & Quality Engineering

Data cleaning routines were applied systematically to ensure the statistical validity of subsequent analytics steps:

- **Missing Value Treatment:** Addressed sparse matrix features via logical statistical imputation and record exclusion protocols.
- **Duplicate Removal:** Purged non-unique primary keys to prevent double-counting variances in metrics.
- **Data Type Conversion & Formatting:** Standardized datetime matrices to ISO formatting and cast numerical values to float types.
- **Outlier Detection:** Executed standard distribution filtering to detect and quarantine computational anomalies.
- **Consistency Checking:** Confirmed cross-variable logic fields (e.g., ensuring zero costs align with 100% discount codes).

Step 3: Advanced Exploratory Data Analysis (EDA)

The exploratory layout assessed structural attributes across multiple analytical depths:

- **Univariate Analysis:** Evaluated single-variable spreads using histograms, frequency distribution matrices, and box-plots to assess centrality and dispersion for *Total Cost*, *Total Items*, *Season*, *Payment Method*, and *Store Type*.
- **Bivariate Analysis:** Examined relationships between categorical variables and continuous outputs, validating correlations like *Discount vs. Total Cost*, *Store Type vs. Aggregate Sales*, and *Seasonal Distribution vs. Total Revenue*.
- **Multivariate Analysis:** Rendered full-scale correlation matrices, heatmaps, pairwise charts, and multi-factor grouping arrays to model the complex dependencies of systemic variables on the store's performance.

Step 4: Interactive Dashboard Architecture

A corporate-grade Power BI dashboard architecture was constructed to give business stakeholders immediate access to performance trends across key operational areas:

- **Core Strategic KPI Blocks:** Institutional tracking for Total Volume, Top-Line Revenue, Average Order Values, and Average Basket Sizes.
- **Analytical Visualizations:** Embedded temporal trendlines for monthly sales tracking, proportional breakdowns of payment/store modalities, and cross-regional comparisons.

KEY METRICS & ANALYTICS FINDINGS

The exploratory model revealed highly uniform performance metrics across the entire enterprise matrix, proving a consistent and predictable retail baseline across all tracking categories:

\$52.46 AVG TRANSACTION COST	5.5 items AVG BASKET SIZE	50.0% DISCOUNT ADOPTION	Balanced SEASONAL REVENUE
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- **Customer Purchases & Invoicing:** Transactions yielded a tight mean distribution of \$52.46 with a predictable baseline cart size of 5.5 units per checkout event.
- **Promotional Distribution:** Promotions and active discounts were identified in exactly 50% of historical transactions, allowing for a balanced split test between baseline and promotional behaviors.
- **Structural Channel Uniformity:** The analysis highlighted an even distribution across payment methods (Cash, Card, Wallet) and customer segments. This indicates highly balanced operational profiles across regions, periods, and store layouts.

CONCLUSION

This analytics engagement successfully fulfills the full-lifecycle data pipeline, transforming over one million raw transactional records into high-value corporate insights. By linking strict data cleaning, multi-stage exploratory statistics, and a centralized Power BI business intelligence interface, the project provides leadership with a clear view of systemic operational data. The flat distributions uncovered suggest a reliable, well-hedged market footprint. This enables corporate leadership to execute strategic expansions, optimize inventory distribution patterns, and fine-tune promotional campaigns with high confidence.